

Raqib Abdullah Muktadir

AI Engineer · Test Automation · Full-Stack Builder

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PROFESSIONAL SUMMARY

AI-oriented engineer with a CS degree and a track record of building automation systems, LLM-powered tools, and full-stack applications that ship to production. At Matrox, spearheaded a test automation initiative that had been stalled for over 2 years — delivering 80% of the target automation coverage within 2 weeks and reducing manual QA cycle time by 30%. Independently shipped 6+ projects spanning AI integrations, live event tooling, and finance apps. Driven by a clear mission: build AI-powered products that create genuine impact.

WORK EXPERIENCE

Test Automation Developer, IPKVM Products | [Matrox Video](#) | Montréal, QC *April 2025 – Present*

- **Spearheaded an Avio2 IP-KVM WebUI test automation initiative that had been stalled for 2+ years** — delivered a working, CI-integrated framework covering 80% of target test scenarios within 2 weeks, authoring 11 Robot Framework test scripts from scratch using a Playwright-based browser library.
- **Reduced manual QA verification effort by 30%** by integrating automated scripts into the Jenkins release pipeline, replacing repetitive human-run regression cycles with headless Chromium-driven test runs triggered on every build.
- **Designed and deployed Jenkins CI/CD pipelines:** a standard parameterized suite and a long-run stress pipeline engineered to sustain 1,000+ loop iterations with per-loop result isolation, orphaned Chromium process cleanup every 50 cycles, and automated HTML email reporting via SMTP after every run.
- **Developed a PyQt5 desktop GUI (Avio2 Device Manager)** — replacing fragmented per-device browser sessions with a single interface providing live REST API data, bitrate/codec controls, NMOS status, and device identity management across multiple simultaneous units.
- **Authored REST API automation using RequestsLibrary across 3 distinct API namespaces (auth, mgmt, app):** Bearer token auth flows, reboot-triggered health polling with up to 3 minutes of recovery monitoring, and full endpoint coverage mapped from internal OpenAPI/YAML specs.
- **Proactively identified and reported a session token security observation (Chrome Session Storage exposure) to the dev team**, following up with XSS protection suggestions — demonstrating security-first engineering mindset beyond QA scope.
- **Solved 10+ non-trivial automation challenges:** self-signed TLS cert handling, JavaScript-injected double-click events, CDK overlay interception, cross-machine Jenkins node setup with dynamic agent selection, and Windows-safe result merging via Python glob.

Software Developer Intern | [Matrox Video](#) | Montréal, QC *May 2024 – April 2025*

- **Improved internal inventory system performance by 20%** by redesigning SQL queries and resolving long-standing data bottlenecks in a PHP/MySQL full-stack developer productivity tool used daily by the QA team.
- **Built a full automated bug-reporting system integrating test logs with the Jira API**, reducing manual documentation time by 30% and eliminating the manual copy-paste workflow between test runs and issue tracking.
- **Built automated CI/CD pipelines in Jenkins for IP and SDI video workflows;** translated client-reported failures into structured, reproducible test cases — measurably reducing defect recurrence across firmware releases.
- **Designed overnight automation frameworks in PowerShell and AutoIt** executing hundreds of nightly test cycles with real-time log generation, surfacing intermittent bugs in hours that previously required multi-day manual investigation.
- **Developed C#-based visual automation scripts** capturing monitor snapshots during FPGA hardware simulations, comparing outputs against expected reference patterns, and logging discrepancies for firmware regression tracking across releases.

Co-Founder & Experience Lead | [555 Studio \(FiveFiveFive\)](#) | Montréal, QC *September 2024 – Present*

- **Co-founded and operate a Montreal-based creative speaker series;** shipped 10+ events with 711+ registrations and a 100% venue fill rate — zero events undersold — with partners including Shopify, ETS, HEC, and Cansbridge.
 - **Designed and built the official 555 Studio website (555-studio.com) end-to-end:** event details, speaker showcases, and registration flow in a cohesive Next.js + TypeScript frontend.
 - **Led speaker outreach, sponsorship negotiations, and end-to-end event operations** across formats including Presentation Night (5 stories / 5 slides / 5 min), Builder Sundays with Shopify, and Electra — a live audio-visual performance experience.
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TECHNICAL PROJECTS

Avio2 WebUI & REST API Automation Suite — *Robot Framework · Playwright · Python · Jenkins · Groovy · RequestsLibrary*

- 11 end-to-end test scripts covering 7 authentication scenarios, session management, tab navigation, device control, bitrate cycling, and API-triggered reboot with automated 3-minute recovery polling.
- Jenkins pipeline with dynamic multi-node agent selection, parameterized device IPs, 10–12 hour timeouts, and automated HTML email delivery — no manual steps from trigger to report.

NewsFlow — AI News Aggregator — *Next.js · TypeScript · Prisma · SQLite · GPT-4o-mini*

- Pulls from 20+ RSS sources (BBC, Reuters, TechCrunch and more), deduplicates via a custom 3-layer pipeline (exact hash → Levenshtein → cosine similarity on word-frequency vectors), and serves stories ranked by a recency + corroboration scoring formula.
- Integrated an independent GPT-4o-mini assistant that searches both the article database and the live web — architecturally isolated from the aggregator pipeline to prevent context contamination.

AI Product Readiness Critic — *TypeScript · Next.js · OpenAI API · Zod*

- Structured LLM critique engine with a modular prompt composition system: base template + injectable slots for strictness level (Gentle / Normal / Brutal) and persona — all model responses Zod-validated before rendering to prevent silent failures.

Spotify Music Discovery AI — *Next.js · TypeScript · OpenAI API · Spotify API*

- Music discovery chatbot that recommends tracks based on natural language mood input, integrating OpenAI for intent parsing and the Spotify API to process real-time user listening data.

AudioVisual — 555(SOUNDS) — *TypeScript · Three.js · Web Audio API*

- Real-time VJ-lite performance tool built for live 555 Studio events: audio-reactive 3D scenes in Three.js driven by Web Audio API analysis, with live cursor control for on-stage performance.

HDMI Visual Test Automation — *C# · FPGA · Computer Vision*

- Drives FPGA-generated HDMI signals, captures live video frames via webcam, and compares against reference patterns to flag discrepancies — a precursor architecture to VLM-based visual grading systems.

EDUCATION

BSc Computer Science, Minor in Business Administration | [Memorial University of Newfoundland](#) | St. John's, NL

SKILLS

Languages	Python · TypeScript · JavaScript · C# · Java · PHP · SQL · Groovy · C++
AI / LLM	OpenAI API · Anthropic Claude API · Prompt Engineering · Structured Outputs · Zod Validation · RAG Architecture · AWS Bedrock (studied) · Claude Code
Frameworks	Next.js · React · Node.js · Robot Framework · Playwright · Selenium · FastAPI · Prisma · PyQt5 · Three.js · AutoIt
Databases	SQLite · PostgreSQL · MySQL · MongoDB · pgvector (studied) · ChromaDB (studied)
DevOps / CI	Jenkins · GitHub Actions · Docker (studying) · Git / GitHub · PowerShell · Jira API · Groovy Pipelines
Tools	VS Code · Postman · Chrome DevTools · Warp Terminal · Claude Code
Human Lang.	English (fluent) · French (professional) · Bengali · Spanish · Hindi / Urdu